

PELLEY BAY, NU, Canada

WMO: 719180

Lat: 68.437N

Lon: 89.726W

Elev: 324

StdP: 97.49

Time Zone: -7.00 (NAM)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-42.0	-40.5	-40.7	0.1	-36.9	-40.4	0.1	-36.8	19.8	-25.7	18.2	-28.9	6.8	270	1.196

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.0	17.3	12.7	15.1	11.0	13.1	9.4	12.6	16.4	10.2	13.6	8.5	11.5	5.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.3	8.1	14.0	7.9	6.9	11.8	6.4	6.2	9.8	36.6	16.4	30.4	13.8	26.7	11.4	20.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.7	15.8	13.8	DB	-42.9	20.4	2.8	2.8	-45.0	22.3	-46.6	23.9	-48.2	25.5	-50.2	27.5	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-13.8	-31.5	-31.7	-27.7	-19.4	-9.7	1.1	7.9	5.2	-1.8	-10.6	-21.0	-27.2
	DBStd	14.91	5.69	5.87	6.84	6.09	4.71	4.68	4.00	4.04	3.81	5.23	6.65	6.51
	HDD10.0	8711	1287	1167	1168	882	611	272	92	158	354	637	931	1152
	HDD18.3	11711	1545	1401	1427	1132	869	517	324	406	604	896	1181	1410
	CDD10.0	41	0	0	0	0	0	4	26	11	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.6	7.0	6.3	6.0	6.6	7.5	6.7	6.2	6.4	6.4	7.2	6.6	6.5
Precipitation	PrecAvg	240	3	3	5	11	21	20	38	53	40	29	9	6
	PrecMax	438	18	17	32	47	68	54	118	141	99	100	33	44
	PrecMin	111	0	0	0	0	0	0	12	8	7	6	0	0
	PrecStd	77	4	4	8	9	15	13	21	32	22	19	8	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-10.0	-14.6	-9.0	-3.2	0.6	18.3	21.1	19.7	11.2	0.9	-4.4	-6.7
		MCWB	-11.2	-15.1	-9.3	-3.5	-0.7	10.2	13.6	15.0	9.3	0.7	-4.6	-7.9
	2%	DB	-17.5	-17.2	-12.2	-6.1	-0.7	14.1	18.0	16.2	8.1	-1.5	-6.8	-11.4
		MCWB	-17.8	-17.7	-12.7	-6.3	-1.2	9.1	13.3	13.1	5.7	-1.8	-7.0	-11.9
	5%	DB	-20.6	-19.9	-14.8	-8.2	-2.0	10.8	16.0	13.6	6.0	-2.8	-9.1	-14.5
		MCWB	-20.9	-20.2	-15.1	-8.6	-2.3	7.4	11.4	10.3	4.8	-3.0	-9.3	-14.8
	10%	DB	-23.1	-22.6	-17.5	-10.3	-3.5	7.8	14.3	11.3	3.6	-4.1	-11.1	-17.7
		MCWB	-23.3	-22.9	-17.9	-10.7	-3.7	5.4	10.0	8.1	3.0	-4.5	-11.4	-17.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-13.1	-15.6	-12.1	-4.6	-0.5	10.7	15.0	15.7	9.7	1.5	-5.6	-10.2
		MCDB	-13.1	-15.3	-11.8	-4.3	-0.4	13.8	18.4	18.9	11.0	1.7	-5.4	-10.0
	2%	WB	-18.5	-18.0	-15.6	-8.0	-1.8	8.7	12.8	13.2	6.8	-2.0	-8.9	-12.4
		MCDB	-18.2	-17.8	-15.4	-7.8	-1.4	12.7	17.5	15.9	8.4	-1.9	-8.7	-12.0
	5%	WB	-22.1	-19.6	-18.5	-10.1	-3.1	7.1	11.0	8.6	4.5	-4.0	-11.5	-15.6
		MCDB	-22.0	-19.3	-18.3	-9.7	-2.9	9.7	15.0	10.8	5.8	-3.9	-11.3	-15.3
	10%	WB	-25.2	-22.2	-20.3	-12.8	-4.6	5.5	9.3	6.8	2.4	-5.5	-15.5	-20.0
		MCDB	-25.1	-21.9	-20.0	-12.4	-4.3	7.5	12.8	8.9	3.0	-5.3	-15.3	-19.8
Mean Daily Temperature Range	5% DB	MDBR	4.8	5.0	5.8	5.9	4.6	4.4	6.0	5.1	3.5	4.0	4.9	4.6
		MCDBR	7.0	6.4	6.8	6.4	5.0	8.1	8.1	8.2	6.4	4.4	6.4	6.5
		MCWBR	9.3	6.2	6.1	5.5	5.6	4.2	4.8	6.9	5.0	5.3	7.6	5.4
	5% WB	MCDBR	9.4	6.3	6.3	5.7	5.6	5.6	7.0	8.3	5.8	4.8	8.0	5.2
		MCWBR	9.3	6.2	6.2	5.8	5.6	4.3	5.2	6.5	4.9	4.8	7.7	5.2
Clear-Sky Solar Irradiance	taub	0.142	0.197	0.243	0.285	0.310	0.308	0.339	0.310	0.282	0.227	0.147	N/A	
	taud	1.893	2.097	2.201	2.136	2.144	2.375	2.415	2.518	2.471	2.104	1.908	N/A	
	Ebn at Noon	136	635	808	859	873	893	847	842	772	548	104	0	
	Edh at Noon	6	43	74	108	122	100	94	76	59	41	5	0	
All-Sky Solar Radiation	RadAvg	0.03	0.45	2.01	4.43	6.13	5.98	5.12	3.46	1.69	0.53	0.08	0.00	
	RadStd	0.00	0.03	0.07	0.15	0.36	0.53	0.51	0.27	0.14	0.06	0.01	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+0.91	+0.95	N/A	N/A	N/A	N/A	-436	-432	N/A	N/A

Nomenclature: See separate page